Discover. Learn. Empower.

Subject: Theory of Computation

Subject Code: 23CST-316/ 23ITT-316

Semester: 6th Batch: 2023

Slow Learner's

Assignment (MST-2)

Total Marks: 10

*Note: Every student who have score less than 10 marks in MST-2 needs to submit Slow Learner assignment.

Q. No.	Question	CO's
1.	Let the production of the grammar be S-> 0B | 1A, A-> 0 | 0S | 1AA, B-> 1|1S |0BB. For the string 0110 find the right most derivation.	CO4
2.	Differentiate between ambiguous and unambiguous grammars and prove that following grammars are ambiguous: a) E->E+E/E*E/(E)/id b) S->SS/(S)/a/∧	CO3
3.	State rules to remove null-productions from CFG. Consider the grammar G with following productions: $S \rightarrow aS \mid AB,\ A \rightarrow \Lambda,\ B \rightarrow \Lambda,\ D \rightarrow b.$ Remove null Productions from this given grammar.	CO4
4.	Convert the given CFG into CNF. $S \rightarrow aAD,\ A \rightarrow aB \mid bAB,$ $B \rightarrow b,\ D \rightarrow d.$	CO4

5.	How can you prove that given language is not regular by Pumping lemma? Discuss various steps.	CO3